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DECISION-RULES AND INDIVIDUAL VALUES IN CONSTITUTIONAL CHOICE*

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Once a political community has decided which of its members are to participate directly in the making of collective policy, an important question remains: "How many of them must agree before a policy is imposed on the community?" Only if participation is limited to one man does this question become trivial. And this choice of *decision-rules* may seem only a little less important than the choice of rules¹ in a world so largely governed by committees, councils, conventions, and legislatures. This paper is about the consequences of these rules for individual values.

Both the oral and written traditions of political theory have generally confined the search for optimal (or "best") decision-rules to three alternatives.² The *rule of consensus* tells us that all direct participants must agree on a policy

which is to be imposed. *Majority-rule* tells us that more than half must concur in a policy if it is to be imposed. And the *rule of individual initiative* (as we may call it), holds that a policy is imposed when any single participant approves of it. These three decision-rules—"everyone," "most of us," and "anyone"—are terribly important, but they cannot be said to exhaust the available alternatives.

The list of alternatives is just as long as a committee's roster. Only for a committee of three would 'consensus,' 'majority' and 'individual initiative' exhaust the possibilities. In a committee of n members, we have n possible rules. Let the decision-rule be a minimum number of individuals (k) required to impose a policy. For a committee of five, we have five alternative values of k :

individual initiative		majority rule		rule of consensus
↓		↓		↓
$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
⏟	⏟	⏟	⏟	⏟
(n = 5)				

* An earlier version of this paper was read at the International Conference on the Mathematical Theory of Committees and Elections, Vienna, June, 1968. For helpful criticisms of that earlier version, I thank Jay Casper, Richard Curtis, Robert Dahl, William Riker, Arnold Rogow, and Michael Taylor.

¹ Well, maybe somewhat more than a "little less important," but not trivial by comparison: for evidence in a very practical argument, consider J. J. Servan-Schreiber, *The American Challenge* (New York: Atheneum, 1968), pp. 170-178.

² This limitation is illustrated by Abraham Lincoln's remark: "Unanimity is impossible; the rule of a minority as a permanent arrangement, is wholly inadmissible; so that, rejecting the majority principle, anarchy or despotism in some form is all that is left." Cited in Henry B. Mayo, *An Introduction to Democratic Theory* (New York: Oxford University Press, 1960), p. 178, and in Thomas Landon Thorson, *The Logic of Democracy* (New York: Hold, Rinehart and Winston, 1962), p. 142.

This means, of course, that the number of possible rules is often very large. The U.S. House of Representatives might, for example, choose any of 435 decision-rules; the British House of Commons could choose any of 630 rules; and the Supreme Soviet might choose any of 1300 rules. A committee of n can, in short, choose any of n possible rules.³ And one can hardly claim to have identified the 'best' decision-rule,

³ Rules with a value equal to or less than n over 2, of course, present special liabilities, for they permit the formation of more than one winning coalition on a single issue. But for some classes of issues this is not a serious difficulty, since multiple policies may be effected. An example is provided by *certiorari* proceedings in the U.S. Supreme Court: in these instances, the so-called "rule of four" ($k = 4$, $n = 9$) is applied, so that any four justices may require that a writ be granted. This produces no difficulties, since the granting of one writ does not preclude the granting of another—multiple policies are permissible.

whatever his normative standard, unless he has shown it superior to the $n-1$ alternatives.

This paper reports one normative analysis of the decision-rules available to a generic political committee.⁴ In it, I consider a single, anonymous individual who wishes to optimize the correspondence between his own values, however selfish or altruistic, and those expressed by collective policy. This individual would like to "have his way" as often as possible, by securing the adoption of proposals he likes and the defeat of proposals he dislikes. For him, then, the "optimal" decision-rule will be the one which minimizes the frequency with which he must expect to support a proposal yet have it defeated, or to oppose a proposal yet have it adopted. This is because his values will correspond to collective policy most closely under rules which minimize the frequency with which he is "outvoted" in one of these two ways. And, for him at least, the best decision-rule will be the one which minimizes the expected frequency for these unhappy events. The resulting normative criterion is elaborated in Part I below.

The formal analysis begins (in Part II) with a simple model for the operation of political committees. This is a "null" model in the sense that it presumes very little information about the future. Under it, we know nothing about the (long-run) agenda which will confront the committee, about the ways individuals will evaluate the proposals which do arise, or about the factional structure of the committee. This model serves two purposes. First, it produces a recommendation appropriate to the circumstances of "constitutional choice": If we presume to know very little about an indefinite future, this model rigorously demonstrates the optimality of majority-rule under our individualist criterion. And second, the results of the simple model provide a "baseline" against which to consider the normative implications of additional information about the future. In Part III, we consider the consequences of information which would, for example, lead us to expect the formation of a standing minority faction, or a general prejudice for or against government action.

Let me begin by explaining my normative criterion and distinguishing it from three leading arguments about decision-rules—those which rest on the ideals of equality, economy, and "truth."

⁴The term "political committee" is used here in its broadest sense—to include all bodies which vote on collective policies. This use is illustrated by Duncan Black's book, *The Theory of Committees and Elections* (Cambridge: Cambridge University Press, 1963).

I. A NORMATIVE CRITERION

Let us suppose that the individual with whom we are concerned has some personal schedule of values. We know nothing about the specific content of his value schedule—its selfishness or altruism, its sophistication or naivete, its generality or specificity—since this individual, whom we may call "Ego," is a purely generic figure. We must, however, suppose that his schedule is complete enough that it leads Ego to like or dislike each proposal which comes before the committee. Accordingly, or major assertion is that Ego would like to optimize the correspondence between his preferences and the policies which are imposed by the committee:⁵ he wants to have his way by defeating proposals which his values lead him to dislike and by imposing those which they lead him to like. Our major normative assumption is then:

*Value Assumption:*⁶ Ego wishes to optimize the correspondence between his schedule of values and the list of policies which are imposed.

Now let us suppose that Ego looks ahead and contemplates four events which might occur:

A. Ego's value schedule leads him to support a proposal, but it is not imposed.

B. Ego's value schedule leads him to oppose a proposal, but it is imposed.

C. Ego's value schedule leads him to support a proposal, and it is imposed.

D. Ego's value schedule leads him to oppose a proposal, and it is not imposed.

Ego need not be much of a logician to see that our "Value Assumption" will be best served by the frequent occurrence of events C and D. Since it is also clear that the four classes of events are disjoint, Ego can be sure that a decision-rule which minimizes the summed frequencies of A and B will also maximize the summed frequencies of C and D. Hence, Ego

⁵This assumption is less innocuous than it may appear: complete indifference is not improbable even among the very active (viz. committee members).

⁶Here, I am putting aside the theoretically important "problem of intensity." In effect, the value assumption implies that all issues evoke equally intense preferences. This decision rests on theoretical intractability of the problem. Some of the major difficulties are outlined in, Willmoore Kendall and George Carey, "The 'Intensity' Problem and Democratic Theory," this *Review*, 62 (1968), 5-24, and in Douglas W. Rae and Michael Taylor, "Some Ambiguities in the Concept of 'Intensity,'" *Polity*, 1 (March, 1969).

can see that the lower the frequency of events A and B, the higher the frequency of C and D, and the higher the degree of correspondence between his schedule of values and the list of policies imposed by the committee. This leads to a criterion by which Ego may evaluate decision-rules.

Criterion for Choice Between Decision-Rules:

One should choose that decision-rule which minimizes the sum of the expected frequencies for (A) in which the committee does not impose a policy which has value schedule leads him to support, and (B) in which the committee imposes a policy which his value schedule leads him to oppose.

This is the normative criterion which underlies our analysis. Each decision-rule will be evaluated by the frequency with which we must, under given assumptions, expect these events to occur if that decision rule is adopted. Our initial commendation of majority rule will imply nothing more than this: under majority rule, the expected (summed) frequency of events A and B is at a minimum, and majority rule is therefore optimal under this criterion. This in turn suggests simply that majority rule maximizes the probability that our (anonymous) individual will "have his way" with respect to a given proposal. And this, in the long run, suggests that majority rule will optimize the correspondence between individual values and collective policies. Since "having one's way" is its central mechanism, we may call this criterion *political individualism*.

The "moral tone" of this position is exactly that of the individuals with whom we are dealing—no better and no worse. Under the criterion of political individualism, we must recommend decision-rules which optimize the correspondence between individual values and collective policies, even if we do not like these values, and even if we do not consider these values "rational" in some larger context.

The best known alternative to this criterion is also independent of the "morality" or "rationality" of individual values: this is, of course, the criterion of *political equality*. While "equality" is a word of many meanings, its most relevant use for present purposes is the rule that, "... the weight assigned to the choice of each individual is identical."⁷ A minimum consequence of this criterion is that all decision-rules must be defined by the *number* of members who must vote yes if a policy is to be imposed, regardless

of *which* members these may happen to be.⁸ This criterion is the best-known basis for recommendations of majority-rule. As Neal Reimer puts the case, any alternative rule amounts "... to ignoring the majority: to declaring that the minority shall be considered and weighted *unequally*."⁹ In the present analysis, this is taken for granted: we are equally concerned with each individual member, and we define decision-rules without respect to the identities of the individuals voting yes and no. Here we are examining the additional requirement that a decision-rule optimize the expected correspondence between these (equally valued) individual preferences and the policies which a committee adopts.

A second and more restrictive alternative to political individualism is offered by the family of criteria based on the *truth-value* of collective policies. An example is provided by the 15th Century Council of Ferrara, which adopted a two-thirds decision-rule for proposals to alter Church doctrine, apparently on the presumption that this was a hedge against theologically fallacious changes.¹⁰ Another example is held out by Rousseau's first criterion for the choice of a decision-rule: "... the more grave and important the questions discussed, the nearer should the opinion that is to prevail approach unanimity."¹¹ The apparent presumption is that voting

⁸ This requirement is closely related to K. O. May's condition of "anonymity," under which the decision-rule does not depend in any way upon the identification of individual members. See his "A Set of Independent Necessary and Sufficient Conditions for Simple Majority Decision," *Econometrica*, 20 (1952), 680-684. May's paper shows that majority rule alone satisfies "anonymity" and three other conditions simultaneously.

⁹ Neal Reimer, "The Case for Bare Majority Rule," *Ethics*, 62 (1951-52), 16-32. The cited statement is from p. 17.

¹⁰ Thomas Baty, "The History of Majority Rule," *The Quarterly Review*, 216 (1912), 1-28. See especially pp. 22-23. The Council also was tri-cameral: within each chamber, a two-thirds rule prevailed; between chambers the rule of consensus was followed. This is indeed a restrictive decision-rule, and a fine hedge against doctrinal error, if we assume the existing doctrine is altogether correct.

¹¹ *The Social Contract*, translated by G. D. H. Cole (New York: E. P. Dutton, 1950), p. 107. Rousseau's second rule was that "... the more the matter in hand calls for speed, the smaller the prescribed difference in the number of votes may be allowed to become: where an instant decision has to be reached, a majority of one should

⁷ See Robert A. Dahl, *A Preface to Democratic Theory* (Chicago: University of Chicago Press, 1956), pp. 67, 84.

is a sort of sampling device for the determination of the General Will: restrictive sampling criteria reduce the chance of a policy inconsistent with the General Will being adopted. Our criterion makes no claim to the optimization of truth values, mainly because such a criterion seems epistemologically indefensible, and because it suggests that past policy is necessarily "truthful," independent of changing circumstances.

Yet a third alternative, much closer to the present criterion, is offered by *economic individualism*. This criterion is embedded in the often fruitful application of economic methods to politics—the works of Olson, and Buchanan and Tullock being important examples.¹² As the latter authors put the criterion, "... individuals will, on the average, choose more rather than less when confronted with the opportunity for choice in the political process, with more and less being defined in terms of measurable economic position."¹³ In their analysis of decision-rules, Buchanan and Tullock define two cost functions: (1) "external costs," which are defined by the costs a single individual "... expects to endure as a result of the actions of others," and (2) "decision-making costs," which are defined by "the time and effort ... which is required to secure agreement ... among the number of people required by a decision-rule to impose a policy."¹⁴ One then looks for a rule which minimizes the sum of these costs.¹⁵

be enough." Condorset actually devised a truth-function theory using probabilistic assumptions similar to those found in Part II below. See *Essai sur l'application de l'analyse à la Probabilité des Décisions Rendues à la Pluralité des Voix* (Paris, 1785).

¹² Mancur Olson, *The Logic of Collective Action* (Cambridge, Massachusetts: Harvard University Press, 1965); and James Buchanan and Gordon Tullock, *The Calculus of Consent* (Ann Arbor: University of Michigan Press, 1962).

¹³ Buchanan and Tullock, *op. cit.*, p. 29. The quotation is one of two interpretations which these authors give to the more general criterion of utility maximization.

¹⁴ *Ibid.*, pp. 64, 68.

¹⁵ *Ibid.*, p. 70. "For a given activity the fully rational individual, at the time of constitutional choice, will try to choose that decision-making rule which will minimize the present value of the expected costs that he must suffer. He will do so by minimizing the sum of the expected external costs and expected decision-making costs. ..." It is not clear that these costs can be added on any single scale. For a commentary on these cost

While this approach is in many respects a powerful one, I have chosen not to follow it here for two major reasons. The first is most important: we are interested in a political problem in its own right, and this must not be submerged in the conceptual (sometimes even ideological) bath of economic individualism. We do not wish, therefore, to conceive the problem of having one's way as a means to success in a larger economic game, but to leave the source of individual values open to the unspecified "value schedules" of individuals. A second, less important reason is methodological: I know of no way to associate costs with decision-rules and at the same time assure that these costs are additive. So, instead, I have chosen to examine expected frequencies for having one's way without subsuming these frequencies under more general cost-functions. While this analysis owes important intellectual debts to these economic models—especially to Buchanan and Tullock—it should not be viewed as a contribution to that tradition.¹⁶

We may now proceed with our analysis, taking political individualism as our normative criterion. We begin with an individual who tries to evaluate decision-rules under the bewildering uncertainty of "constitutional choice."

II. A SIMPLE CONSTITUTIONAL CHOICE MODEL

We are now interested in the frequencies with which our generic number (Ego) must expect the events A and B to occur under a given decision-rule. We begin, here, with a model which generates these expected frequencies from the temporal (and epistemological) perspective of "constitutional choice." As James Buchanan ex-

functions see, Herbert J. Kiesling, "Potential Costs of Alternative Decision-Making Rules," *Public Choice*, 4 (1968), 49-66.

¹⁶ Although Buchanan and Tullock produce no rigorous evaluations of decision-rules, two of their speculations seem especially relevant to the present analysis. First, they point out that majority-rule holds no special *a priori* promise as a cost minimization (p. 81). Second, they suggest that "... the low-cost point on the aggregate 'cost curve' would tend to be represented by simple majority voting," under circumstances which appear comparable to those presented in Part II of this paper (p. 129). No strict demonstration of this tendency is offered, and care must be taken not to confuse the frequency curves employed here with the cost-functions which these authors consider. If any determinate relation exists, and I am not sure one does, it is between the frequencies used here and "external costs" in their analysis.

plains, this perspective drastically limits the information which may be incorporated in our model:¹⁷

The center of attention becomes the mental calculus of the individual as he is confronted with a choice among alternative rules for reaching subsequent political decisions. The individual does not know, nor is he able to predict, what particular issues will be presented subsequent to the adoption of the rule. And, even if he [could] predict with some accuracy the sort of issues that may arise, he could hardly predict his own position vis-a-vis the other members of the group.

The model presented here assumes no knowledge of an indefinite agenda and the preferences of individuals at discrete points over that agenda.

The more formal model developed below rests on six rather concrete assertions which may be stated at the outset:

1. We are dealing with a political committee, that is, one which is authorized to make collective policies.
2. This committee will deal with a succession of policy proposals, and is free to impose or reject each.
3. This list of proposals (agenda) is unknown at the time the decision-rule is chosen.
4. Since the proposals are unknown, so are the preferences of the members on each proposal.
5. The committee will contain n coequal members, and n is at least three ($n \geq 3$).
6. Each member either supports or opposes each proposal.

We suppose that one member of this committee, whom we call Ego, is interested in making a choice among the available decision-rules. His choice may be established by a means-ends inference from: (1) the normative criterion developed above, and (2) the model developed here for the purpose of predicting the frequencies of events A and B under each of the available decision-rules.

Our model assigns expected frequencies to events A and B, and associates these with each value of k in a committee of n members. This is accomplished by making certain assumptions about the way individual committee members decide whether to support or oppose each proposal. These assumptions allow us to generate a

distribution of "support sets" for policy proposals, and these sets may be partitioned by two criteria: (1) their size (do they equal or exceed k ?) and (2) their inclusiveness (how many of them of any size will include or exclude our generalized member, Ego?). These partitions are used to define "sets of support sets" corresponding to events A and B. We are then able to generate expected frequencies for these sets—that is, for the occurrence of events A and B at any point in the committee's agenda. This allows us to compare the expected frequencies of these undesirable events for each available decision-rule (i.e., each value of k), and to apply our criterion of choice to these expectations. Let us begin with our assumptions about individual decision-making by committee members.

Assumptions about Individual Decisions. As we have already indicated, each member either supports or opposes each proposal as it occurs. We know nothing of the proposals (or the member's reactions to them), so we must make certain assumptions about the ways in which members decide whether to support or oppose each proposal. Briefly, we assume that each member decides whether to support or oppose each proposal independently, on the basis of his own value schedule, and that he is exactly as likely as not to support each proposal.

Assumption I: Each member decides whether to support or oppose each proposal on the basis of his own value schedule. Value schedules are independently constructed. Therefore, the probability that one member will support (or oppose) a proposal is independent of that probability for any other member.

Let: P represent probability

E represent the eventuality that member (E) supports a proposal

e represent the eventuality that member (E) opposes a proposal

X, x represent these eventualities for member (X)

$[a/b]$ should be read "a, given b"

Then:

$$P[E/X] = P_E$$

$$P[E/x] = P_E$$

$$P[e/x] = P_e$$

$$P[e/X] = P_e$$

Since "Ego" is a generalized member, Assumption I was already implicit in our Value Assumption (above). Every member is trying to optimize the correspondence between the list of policies imposed and his own, independently

¹⁷ "An Individualistic Theory of Political Process," in David Easton (ed.), *Varieties of Political Theory* (Englewood Cliffs: Prentice-Hall, 1966), pp. 25-37, quotation from p. 29. For a related statement, see also Elinor Ostrom, "Constitutional Decision-Making," unpublished paper.

constructed value schedule. It follows that the decisions of members are independent events, whose probabilities are not causally related to each other. This assumption is not, of course, meant as a descriptive generalization suggesting that members never influence or lead each other: it is simply a model assumption.

Assumption II: The probability that each member will support any proposal is exactly one-half, and, since he must either support or oppose each proposal, the probability that he will oppose any proposal is also exactly one-half.

Let: the notation remain the same.

Then:

$$P_E = \frac{1}{2} = .5$$
$$P_o = 1 - P_E = 1 - .5 = .5$$

This is arbitrary, but there are three reasons not to make any assumption different from Assumption II. First, no other assumption seems more "realistic" in an intuitive sense, and some seem a good deal less "realistic." That, however, is not a major concern. Second, any asymmetrical assumption would enormously complicate our model and its predictions because it would deny us the use of the very simple calculus of probabilities for combinations of equally likely, independent events. Third, and crucially, it can be shown that no alternative to Assumption II alters the location of the minimum value for the expected summed frequencies of events A and B. Our finding is general to the whole set of alternative assumptions. This is demonstrated in the appendix. So, in the absence of any expected alteration in our major prediction, there seems no reason to complicate the model.

Expected Frequencies of Support Sets. All proposals will evoke a response from each member, and, taken collectively, these responses will form a combination. We need to know the number of different combinations with various properties, and the probability for each. One essentially trivial assumption must be added before these calculations are begun:

Assumption III: Proposals do not enter the committee's agenda unless at least one member supports each of them. Therefore, there will be no combinations of responses in which no member supports the proposal.

Let: [n letters] form a combination of responses to a proposal

Other notation remains as above.

Then:

$$P_{\{xyz \dots n\}} = 0$$

This assumption, which has no substantial effect on our major prediction,¹⁸ is simply our way to account for "non-events" in which no member makes a proposal. Given these three assumptions (and elementary probability theory), two important deductions appear:

*Deduction I.*¹⁹ The total number of possible discrete combinations of responses for the committee of *n* members will be two to the *n*th power, minus one:

Let: *n* represent the number of members
N represent the number of possible combinations of responses to any proposal

Then:

$$N = 2^n - 1$$

For example, a committee of three generates seven possible response combinations:

XYZ
XYz
XyZ
xYZ
Xyz
xYz
xyZ

$$\text{And, } 2^3 - 1 = 8 - 1 = 7$$

Our strategy will be to determine what proportion of this number of possible response combinations (later called support sets) are equivalent to events A and B. This, with the help of Deduction II, will allow us to arrive at predicted frequencies for these events under various decision-rules. In passing, it is worth noting that the number of possible combinations expands very rapidly:

n	N
3	7
10	1,023
20	1,048,575
100	12,676 × 10 ²⁶ apx

¹⁸ The effect of this assumption is to discard an event in which every member "has his way" since all would vote no (i.e., Event D.) This simply makes the prediction slightly less optimistic than it would otherwise have been, but it has no effect on the location of the A + B minimum in which we are interested.

¹⁹ This follows from the rule of combinations: the number of combinations of size *f* which may be drawn from *n* objects equals $n!/f!(n-f)!$. If we allow *f* to take all values 0 . . . *n*, the resulting sum is 2^{*n*}. Assumption III eliminates the one case in which *f* = 0 (i.e. $n!/n! = 1$), so the result is 2^{*n*} - 1.

This means, of course, that any effort to literally "count" response combinations (or support sets) of various types will be quite futile.

*Deduction II:*²⁰ Every possible response combination is equally probable, and that probability is always the reciprocal of the number of possible combinations for a committee of n members.

Let: $n = 2$

The notation remains as above.

Then:

$$P_{XEX} = P_{Xe} = P_{eX} = \frac{1}{2^n - 1} = \frac{1}{2^2 - 1} = \frac{1}{3}$$

Since this deduction is general to all values of n , it may, for example, be calculated that the probability for each combination in a committee of ten is one over 1,023.

Now, we may convert what we have been calling "response combinations" into what we will call *support sets*, which contain only the members of any combination who support a proposal. This differs from a combination only in the matter of notation. We drop those who oppose a policy, so response combination XYz becomes support set XY . This does not mean that opposition is ignored, since (Z) 's absence here tells us of his decision quite as much as the inclusion of a small "z" would. And, since no transformation has been made, we can be sure that what we have deduced for response combinations will hold also for support sets. This is so because every response combination corresponds to one and only one support set:

Response Combination	Support Set
XYZ	XYZ
XYz	XY
XyZ	XZ
xYZ	YZ
Xyz	X
xYz	Y
xyZ	Z

Since assumption III eliminates uniformly negative combinations (i.e., xyz), this is general and no empty support sets occur.

Expected Frequencies for Sets of Support Sets Corresponding to Events A and B. We may now define events A and B by the "sets of sets"—

²⁰ This follows from Assumptions I and II. The combinations are compounded of independent, equally probable events (i.e., individual decisions), and are therefore themselves equally probable events. Since there are $2^n - 1$ combinations (Deduction I), and each is equally probable, each has a probability of $1/2^n - 1$

collections of support sets—to which each corresponds, and compute the expected frequency of each. (A) Event A occurs when our generalized member (Ego) supports a proposal (belongs to a support set) but is unable to impose it (the size of the set is less than the decision-rule value, k). We may identify the set of support sets (A) which corresponds to event A:

Let: A be the set of support sets which defines the occurrence of event A.

k be the minimum number of members required by the decision-rule to impose a policy.

S be a support set.

S_f be a support set of size f .

$\in$ should be read "lies in"

$\approx$ should be read "is equivalent to"

Then:

$$A \approx [S_f, \text{ such that, } f < k, \text{ \& Ego } \in S]$$

This says simply that the set of sets (A) which corresponds to event A contains all those support sets of sizes (f) less than the minimum value imposed under a decision-rule (k), which also include our generalized member. In short, Ego fails to impose a policy which his value schedule leads him to support (event A).

We are now interested in finding out how many support sets belong to set of sets A. This will depend on the decision-rule value (k) for each committee of n members. First, it is apparent that whenever the value of k is one (i.e., the rule of individual initiative), A must be an empty set. Under rule $k = 1$, the mere inclusion of our generalized individual defines the set as $\geq k$ and means that f cannot have a value of less than k . In short:

$$\text{If } k = 1, A \text{ empty}$$

This means that the rule of individual initiative prevents any member from having a proposal he supports defeated (event A). This, however, is not much help with the whole range of values for k which exceed one. A more general formula may be used to determine the number of support sets belonging to A for all values of k .²¹

²¹ This formula is derived from the rule of combinations and our set definitions. Let us begin with the rule of combinations: The number of support sets of size f is:

$$\frac{n!}{f!(n - f)!}$$

Now, we wish to find the number of combinations of any size (f) which contain a generalized member (Ego), in accordance with our set definition.

Let: N_A be the number of support sets in A
Other notation remains as above.

Then:

$$N_A = \sum_{f=1}^{k-1} \frac{(n-1)!}{(f-1)!(n-f)!}$$

This formula is designed to "count" the support

We may accomplish this by saying "If Ego belongs to a set of size f , there will remain $n-1$ members to be combined in a set of size $f-1$ in order to bring our set to size f with Ego included." Accordingly, we subtract 1 from n and f in the combinatory formula, obtaining:

$$\frac{(n-1)!}{(f-1)![(n-1)-(f-1)]!}$$

This may be simplified by letting the -1 terms in the lower right hand bracket cancel each other:

$$\frac{(n-1)!}{(f-1)!(n-f)!}$$

This tells us how many support sets of size f will contain Ego. Now, we also need to know the sum of these values for all instances in which the set's size (f) is less than k . This is accomplished by letting the values be summed from $f=1$ to $f=k-1$. This accounts for summation term:

$$\sum_{f=1}^{k-1}$$

And the result is the formula given:

$$N_A = \sum_{f=1}^{k-1} \frac{(n-1)!}{(f-1)!(n-f)!}$$

sets of size (f) which contain one generalized member (Ego), and to sum those for every value of f up to $k-1$. This tells us how many distinct support sets include Ego and still don't win under any rule k . By dividing by the total number of possible support sets—winning, losing, with Ego or without Ego—we may generalize this formula to the expected frequency of Event A:

Expectation for Event A: Any member of a political committee of size n , operating under decision-rule k , must expect that a proposal he supports will be defeated in:

$$\frac{N_A}{2^n - 1}$$

proportion of the instances in which a proposal comes to a vote. The full formula may be stated:

Let: E_A represent the expected frequency of event A

Other notation as above.

Then:

$$E_A = \frac{\sum_{f=1}^{k-1} \frac{(n-1)!}{(f-1)!(n-f)!}}{2^n - 1}$$

This statement is fully general to committees of any size, and it applies equally to each member of any such committee.

Table 1 gives the expected frequency for event A (E_A) for committees for three to twelve members, and decision rules from 1 to n . We

TABLE 1. EXPECTED FREQUENCIES FOR EVENT A (E_A)
(REJECTION OF SUPPORTED POLICY)*

$n \backslash k$	3	4	5	6	7	8	9	10	11	12
1	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00
2	<u>.14</u>	.07	.03	.02	.01	.00 +	.00 +	.00 +	.00 +	.00 +
3	.43	<u>.27</u>	<u>.16</u>	.10	.06	.03	.02	.01	.01 -	.00 +
4		.47	.36	<u>.25</u>	<u>.17</u>	.11	.07	.05	.03	.02
5			.48	.41	.33	<u>.25</u>	<u>.18</u>	.13	.09	.06
6				.49	.45	.35	.32	<u>.25</u>	<u>.19</u>	.14
7					.50 -	.47	.43	.37	.31	<u>.25</u>
8						.50 -	.48	.46	.41	.36
9							.50 -	.49	.47	.44
10								.50 -	.50 -	.49
11									.50 -	.50 -
12										.50 -

* To associate values of E_A with values of k for a committee of size n , simply read down the vertical columns of the table. The table is asymmetrical since intersects below the diagonal are impossible ($k > n$). Rule of consensus values appear just on the diagonal ($k = n$). Rule of individual initiative values are in the top row ($k = 1$), and majority rule values ($k > n/2$) are underlined.

will postpone our analysis of these expectations until they can be combined with those for B (E_B).

(B) Event B occurs whenever our generalized member (Ego) opposes a proposal (is *not* a member of its support set) and the proposal is nonetheless imposed (the set contains at least k members). We may now identify the set of sets which corresponds to event B:

Let: B represent the set of support sets which defines the occurrence of event B

Other notation remains as above.

Then:

$$B \approx [S, \text{ such that } f \geq k \text{ and } \text{Ego} \notin S]$$

This says simply that set of sets B contains all those support sets which are large enough to impose a policy under decision-rule k and which do not contain our generalized member (Ego). In short, Ego fails to block a policy proposal which his value schedule leads him to oppose.

Under the rule of consensus ($k = n$), set B must be empty, since no support set can equal or exceed k unless it contains our generalized member:

If $k = n$, B empty

For values of k below n , however, a somewhat more complex "counting" problem arises. It is resolved by the formula:²²

²² This formula is also derived from the combination rule. In this instance, we wish to "count" the support sets of size f which do not include Ego and for which f is at least k . We may begin by saying, "If Ego is not to be included in a set of size f , there will be $n - 1$ members to be combined in sets of size f ." We therefore subtract one from n , and leave f unaltered, obtaining:

$$\frac{(n - 1)!}{f![(n - 1) - f]!}$$

This gives the number of sets of size f which exclude Ego. We wish to sum the value of this fraction for all values of f at or above the value of k . Hence, the summation term:

$$\sum_{f=k}^{n-1}$$

The upper limit is set at $n - 1$ for two reasons: (1) we know that no sets belonging to B may be of size n , and (2) this extreme case can produce a minus factorial term, which is nonsensical. The result is the formula given:

$$N_B = \sum_{f=k}^{n-1} \frac{(n - 1)!}{f![(n - 1) - f]!}$$

Let: N_B represent the number of support sets in B.

Other notation remains as above.

Then:

$$N_B = \sum_{f=k}^{n-1} \frac{(n - 1)!}{f![(n - 1) - f]!}$$

(If $f = n$, $N_B = 0$)

This formula "counts" the number of support sets which are at least large enough to impose a policy under any given decision-rule value (k) and which do not include Ego. This tells how many support sets correspond to event B, in which Ego's value schedule leads him to oppose a policy, yet it is imposed. If we divide this by the total number of support sets in a committee of n members, we may generalize it to the expected frequency for the occurrence of event B:

Expectation for Event B: Any member of a political committee of size n , operating under decision rule k , must expect that a proposal which he opposes will be imposed in:

$$\frac{N_B}{2^n - 1}$$

proportion of the instances in which a proposal comes to a vote. The full formula may be stated:

Let: E_B represent the expected frequency of event B.

Other notation as above.

$$E_B = \frac{\sum_{f=k}^{n-1} \frac{(n - 1)!}{f![(n - 1) - f]!}}{2^n - 1}$$

[$f = n$, $E_B = 0$]

Like its counterpart for event A, this statement is fully general. Table 2 gives the expected frequencies for event B (E_B) in committees of three to twelve members and decision-rules (k) of 1 to n .

It may be worth pausing to examine the shapes of the curves for the expected frequencies of events A and B (see Figure 1). Both are *monotonic*: the direction in which expected frequencies for events A and B change with k in a committee of any size does not change. The curves have no "humps" or "dips." Second, the expected frequencies for events A and B are *curvilinear*: the rate at which they change with k varies over the values of k . Notice that the rate of change for each curve is greatest around the middle of the k values (i.e., where $k = n/2$). This means, for example, that the expected frequency of event A increases most

TABLE 2. EXPECTED FREQUENCIES FOR EVENT B (E_B)
(IMPOSITION OF OPPOSED POLICY)

$k \backslash n$	3	4	5	6	7	8	9	10	11	12
1	.43	.47	.48	.49	.50 -	.50 -	.50 -	.50 -	.50 -	.50 -
2	<u>.14</u>	.27	.36	.41	.45	.47	.48	.49	.50 -	.50 -
3	.00	<u>.07</u>	.16	.25	.33	.35	.43	.46	.47	.48
4		.00	.03	<u>.10</u>	<u>.17</u>	.25	.32	.37	.41	.44
5			.00	.02	.06	<u>.11</u>	<u>.18</u>	.25	.31	.36
6				.00	.01	.03	.07	<u>.13</u>	<u>.19</u>	.25
7					.00	.00 +	.02	.05	.09	<u>.14</u>
8						.00	.00 +	.01	.03	.06
9							.00	.00 +	.01	.02
10								.00	.00 +	.00 +
11									.00	.00 +
12										.00

sharply as the decision-rule rises from slightly below $n/2$ to slightly above it, and that the rate of change is slowest near the extreme values of k , 1 and n . Third, the expected frequencies for the two events form curves which are perfect mirror images of each other; as intuition would lead one to expect, the curves are *opposed*. Last, both curves vary between *limits of zero and one-half*: no decision rule (k) makes either event more than an "even bet."²³ And, since no event can be both A and B—the two corresponding sets are disjoint—the summed frequency can never exceed half. Let us turn to the summed expected frequencies and an application of the choice criterion.

Evaluation of Decision-Rules under Constitutional Choice Model. We began by looking for a decision-rule which would optimize the expected correspondence between individual values and collective choices. Under such an optimal rule, a generalized member would expect to have a favored proposal defeated or a disfavored proposal imposed less often than under any other rule. This line of reasoning leads us to the choice criterion:

One should choose that decision-rule which minimizes the sum of the expected frequencies for (A) in which the committee does not impose a policy which his value schedule leads him to support, and (B) in which the committee imposes a

²³ This upper limit occurs because there is only a .5 probability that Ego will vote yes (or that he will vote no.) These expected frequencies must not, then, be confused with the conditional probability that A or B will occur once Ego has voted yes or no.

policy which his value schedule leads him to oppose.

Our model generates expected frequencies which confirm our intuitive suspicion that no location of the decision-rule will minimize the expected frequencies for both events A and B taken singly. This is because the two curves are opposed; as the expected frequency for A goes down, the expected frequency for B goes up, and *vice versa*. Indeed, the minimum for B is the maximum for A, and the minimum for A is the maximum for B. Under the rule of consensus ($k = n$), one can be sure that no proposal he dislikes will be imposed ($E_B = 0$), but he must also expect that his own proposals will often be defeated ($E_A = .5$). Under the rule of individual initiative ($k = 1$), our member can be sure that proposals he supports will always pass ($E_A = 0$), but he must also expect that many he opposes will also pass ($E_B = .5$). He cannot avoid being outvoted by choosing an extreme decision-rule.

But, as we said, the object is to minimize the sum of the expected frequencies for these two events. The optimal decision-rule must then be selected on the basis of the summed expected frequencies for events A and B. The 'best' rule will correspond to the point at which the sum of these frequencies is minimized for a committee of size n . Table 3 gives the summed frequencies in which we are interested.

The outcome of our search for an optimal decision-rule is apparent in these summed values: *majority rule corresponds uniquely to the minimum summed frequency of events A and B in committees with an odd number of members, and shares the minimum with the value n over 2*

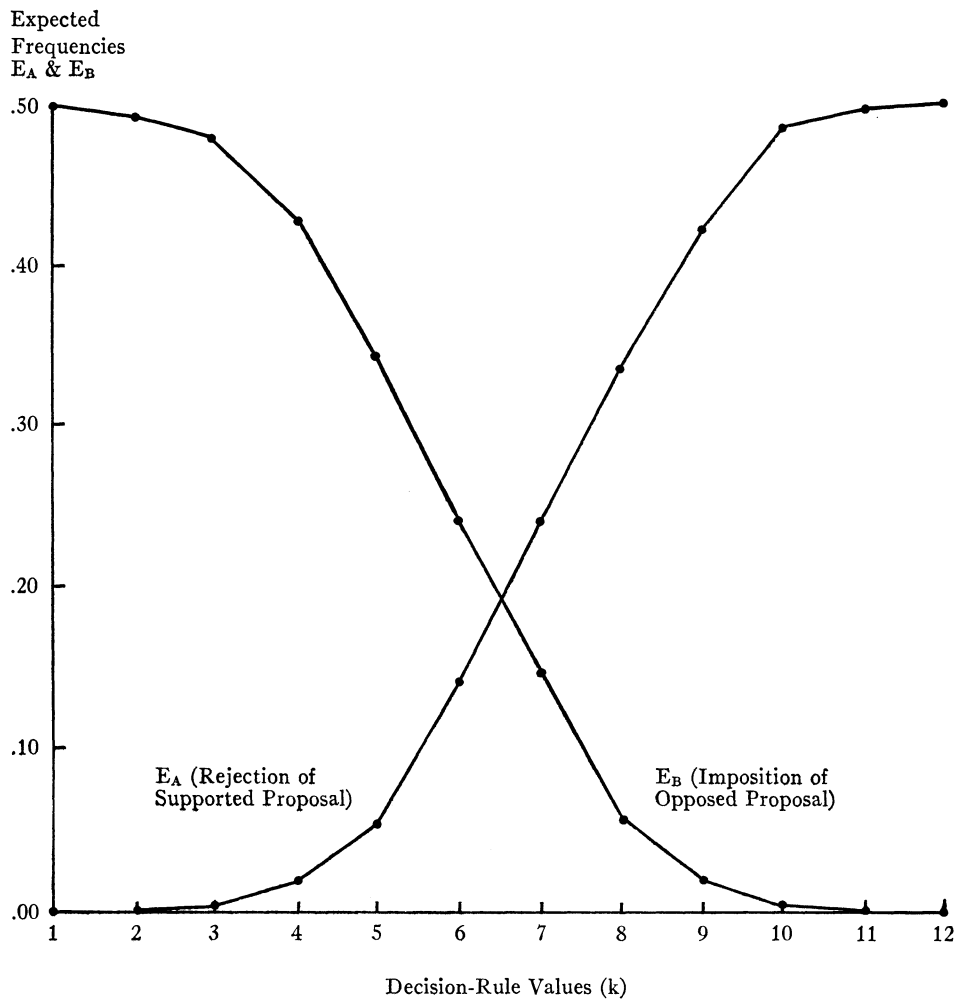


FIG. 1. Expected Frequencies for Events A and B ($n = 12, k = 1 \dots 12$).

NOTE: These expected frequencies apply to long agenda, and may be equated with probabilities for single items on agenda. The reader may use the tables to draw curves for other values of n and k .

TABLE 3. EXPECTED SUMMED FREQUENCIES FOR EVENTS A AND B

<i>n</i> <i>k</i>	3	4	5	6	7	8	9	10	11	12
1	.43	.47	.48-	.49-	.50-	.50-	.50-	.50-	.50-	.50-
2	.28	.34	.39	.43	.46	.47+	.48+	.49+	.50-	.50-
3	.43	.34	.32	.35	.39	.38	.45	.47	.48	.48+
4		.47	.39	.35	.34	.36	.39	.42	.44	.46
5			.48	.43	.39	.36	.36	.38	.40	.42
6				.49	.46	.38	.39	.38	.38	.39
7					.50-	.47+	.45	.42	.40	.39
8						.50-	.48+	.47	.44	.42
9							.50-	.49+	.48	.46
10								.50-	.48-	.48+
11									.50-	.50-
12										.50-

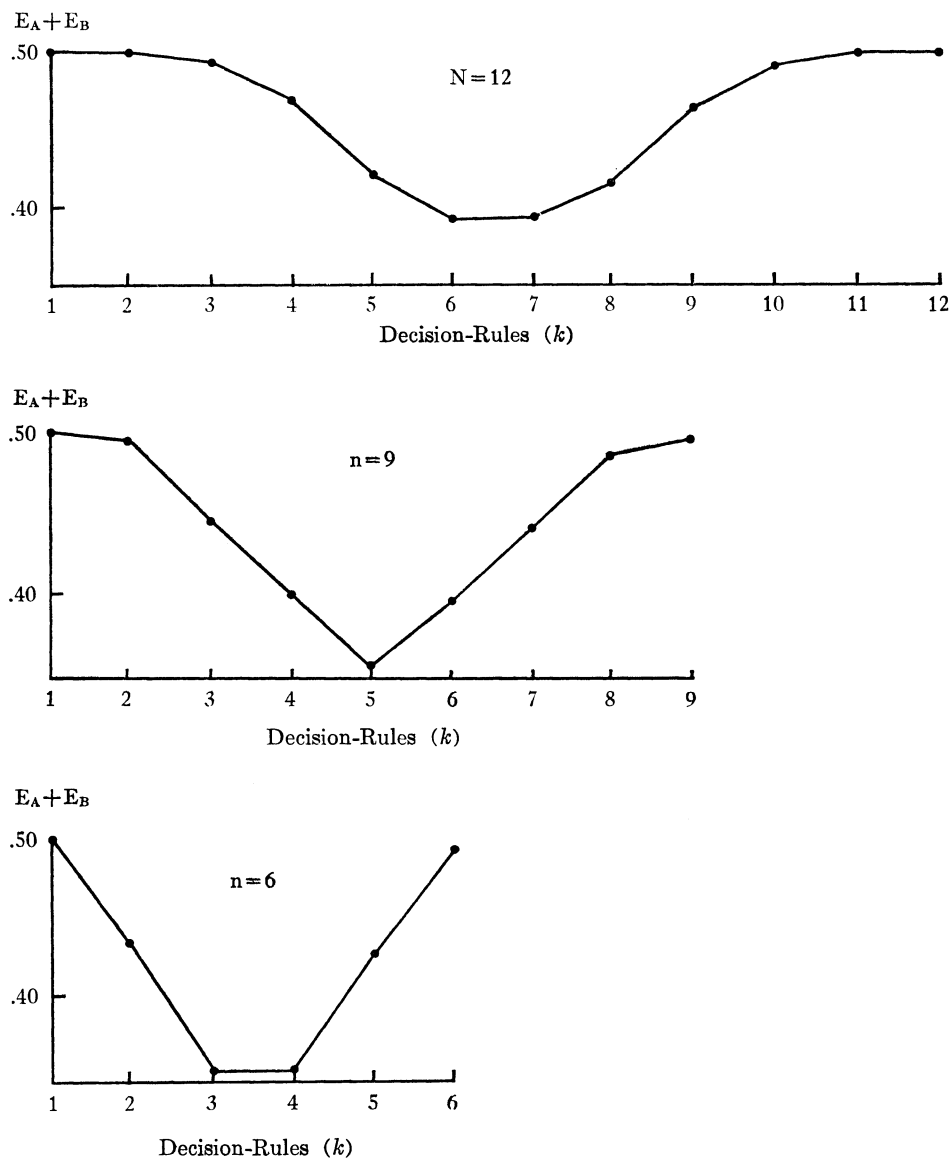


FIG. 2. Expected Summed Frequencies for Events A and B (n as indicated, $k = 1 \dots n$).

in committees with an even number of members. If, under our model, one applies the choice criterion to the range of possible decision-rules, he will be compelled to plump for majority-rule or an adjacent value in every instance. As one proceeds toward the rule of consensus or toward the rule of individual initiative, the expected summed frequencies for events A and B increase. These relations may be seen graphically in Figure 2, and our findings may be stated as follows:

Proposition 1: In committees with an odd num-

ber of members, the optimal decision-rule is majority-rule: n over two, plus one-half.

$$k \text{ [optimal]} = \frac{n}{2} + \frac{1}{2}, \text{ or } \frac{n+1}{2}$$

Proposition 2: In committees with an even number of members, the optimal decision-rule is either majority-rule, or majority-rule minus one: n over two plus one, or simply n over 2.

$$k \text{ [optimal]} = \frac{n}{2} + 1, \text{ or } \frac{n}{2}$$

In so far as the model holds, these proposi-

tions are general, and our choice criterion requires that we select one of the values specified. If one confronts a choice similar to the one confronted in our model by Ego, and if he accepts the major tenets of the model itself, he will find no k value which improves on majority rule:

Proposition 3: In any political committee, majority-rule is as good (i.e., optimal) as any alternative decision-rule, given the model proposed here.

This assertion is, in a sense, more general than existing arguments for majority-rule, since it rests on a comparison with all of its alternatives. And, since the value premise is only an open preference for an optimal correspondence between individual values and collective choices, whatever the individual values may be, the argument may be somewhat less brittle than those which rest on more dogmatic and overarching premises.

The recommendation is also somewhat more general than our (admittedly restrictive) assumptions suggest. These assumptions are meant to represent the problem of "constitutional choice," not to maximize the generality of our conclusion. *Majority-rule is the one decision-rule which precludes the possibility that more people will be outvoted by less people.*²⁴ This accounts for the present conclusion, and it drives us to comparable judgement under any set of assumptions, so long as we can speak of a generic individual²⁵ (as we cannot in the "standing minority" problem considered in Part III below.) This generality is illustrated by the appendix to this paper and is proved by Michael Taylor in a forthcoming article.²⁶ It may, however, be more interesting to consider two sets of assumptions under which our conclusion does *not* apply.

III. SOME ALTERNATIVES

While this recommendation has a certain power of its own, we must remember that "models are models," and that we have quite consciously put aside important aspects of experience. The pure case of constitutional choice

²⁴ This is because all more restrictive rules allow negative minorities to outvote positive majorities, while less restrictive rules allow positive minorities to outvote negative majorities. Under all such rules, it is therefore possible that Event A or Event B will occur for more than half the members on a single vote; under majority-rule (and under $k = n/2$ in committees of even size) this is not possible.

²⁵ And so long as we do not consider "positional preferences," as discussed in Part III below.

²⁶ "A Proof for Rae's Majority-Rule Theorem," forthcoming, *Behavioral Science* (Summer, 1969).

is probably a very rare empirical event, for the choosers of a decision-rule are apt to at least think they know enough about the future to make more determinate guesses than the ones we have made in our model. Let us examine some of the implications which follow from added knowledge about the future.

Let us suppose, first, that a member entertains a *positional* (as opposed to substantive) *preference*, which leads him to think bad actions (Event B) are worse than bad inactions (Event A). This might be so, for example, if he were a "conservative" in Samuel Huntington's sense of that term;²⁷ or if there were some *a priori* difference in the consequences of action and inaction, as with criminal juries in which Ego might consider a bad conviction worse than an objectionable acquittal. And, second, let us suppose that we may anticipate the formation of *factions* whose members possess identical preferences over some class of issues. If this is so, we must suppose that a "permanent minority" may arise and be exploited by a permanent (or rotating) majority. In the following paragraphs, we consider the effects of these guesses about the future on our recommendation for majority-rule.

Positional Preferences. Let us suppose that a member considers it worse for the committee to act against his wishes (Event B) than for it to refuse action in accordance with his wishes (Event A). It appears, for example, that a majority of delegates (including the articulate Madison) at the Philadelphia convention thought it would be worse if Congress expelled a member over a substantial number of objections than if it failed to do so under comparable circumstances.²⁸ Beliefs of this kind—usually related to specific classes of proposals—lead us to assign different "weights" to Events A and B. We may begin thinking about the consequences of these beliefs by defining the admittedly arbitrary quantities which correspond to these

²⁷ In "Conservatism as an Ideology," this REVIEW, Vol. 51, Huntington describes conservatism as a "positional ideology," leading its proponent to an "... articulate, systematic, theoretical resistance to change." (p. 461.) This suggests that a conservative would be apt to weight Event B (unwanted change) more heavily than Event A (unwanted stability).

²⁸ This is an extrapolation from James Madison's *Journal of the Federal Convention*, Vol. 2 (Chicago: Albert Scott and Co., 1893) p. 500. Since the premises of the argument are inexplicit, one cannot be sure that any interpretation is historically correct; the interpretation offered here is, however, at least as plausible as any alternative.

weights:

- 1. Let α represent the weight attached by an individual to Event A, failure to impose a proposal he likes.
- 2. Let β represent the weight attached by an individual to Event B, adoption of a proposal he dislikes.

We must note that these weights are conceptually distinct from the individual's preference on a given proposal. They rest instead on the assertion that, for all proposals within some class, an A outcome may be assigned a weight distinct from that assigned to a B outcome. Given these weights, our criterion of choice is no longer dependent upon the simple sum of the expected frequencies for Events A and B, but on the weighted sum of these frequencies. This revised value assumption recommends the decision-rule which minimizes the value of the weighted frequency:

The optimal rule minimizes: $\alpha \cdot E_A + \beta \cdot E_B$

Drawing the expected frequencies from our simple model (or some other model) and the weights by "flat" from our hypothetical individual, we may now reevaluate a given set of possible decision-rules. Putting aside the enormous difficulty of supplying meaningful quantities for α and β , let us illustrate this alteration of our calculations for a seven-member committee. In this example, we assign weights of one and two to α and β . The question then becomes, "Do these weights justify the adoption of a decision-rule more restrictive than the majority?" On the assumption that the weights themselves make sense, we may answer this question with the weighted frequency term set forth above. In our example, we may answer this question by calculating this term for $n = 7$, $k = 1 \dots 7$, $\alpha = 1$, and $\beta = 2$. The results are shown in the right-hand column of Table 4.

We see that a two-thirds (i.e., $k = 5$) rule is to be preferred in this instance over the simple majority (i.e., $k = 4$) and over all other alternatives. This illustrates the plausibility of restrictive rules where we can say that bad action is worse than bad inaction, and the "calcula-

tions" may very well be analogous to those which underlie the argument for such rules in, say, criminal juries or in bodies which decide upon proposals for constitutional revision. This is not to say that actors have literally made calculations like those given here, but rather that they may have thought the question out in less formal terms and relations similar to those considered here.

This brings us to the suggestion that the simple model presented in Part II is a special case of this more general model: it is the case in which α equals β . In a formal sense, this is self-evident, but we should remember that it is an especially interesting case, for the assumption of equal weights is in general more plausible than any alternative. This is especially evident for political committees whose agenda are open to all types of proposals and which confront dynamic environments which must be expected to undermine any special argument for the *status quo*. In the absence of special evidence to the contrary—like that cited for juries—it seems prudent to accept the equal-weighting case.

Factions. Let us put the problem of factions as bluntly as possible.²⁹ Suppose that we know enough about the committee and its long-term agenda to say that there will, for some class of proposals, be two factions, one of which will contain a majority and one of which will contain a minority of the members. And, suppose further, that we can expect a conflict of interest³⁰ between these factions for some class of issues, so that their preferences will be opposed on each proposal within that class. If we assume all this—as we might in some important cases—then we confront a bald case of the classic difficulty in majoritarian democracy: the exploitation of standing minorities by standing majorities. In our model, Assumption I sheltered us from this prospect, and we may now consider its implications briefly. To begin with, say that:

- 1. A majority faction contains q members ($q > n/2$).
- 2. A minority faction contains $n - q$ members ($< n/2$).

²⁹ The following example states the problem in the simplest possible way, and it is just one of many sorts of examples which might be devised. If one considers more than two factions, and partial degrees of conflict, the argument is vastly more complex, but the essential conclusion of this section is probably unaffected: the prospect of factional exploitation is not a unique attribute of majority-rule.

³⁰ One useful analysis of this concept is offered by Robert Axelrod, *Conflict of Interest: Theory and Political Applications* (unpublished Ph.D. thesis, Yale University, 1968).

TABLE IV. "WEIGHTED" FREQUENCY CALCULATION
($n = 7$, $k = 1 \dots 7$, $\alpha = 1$, $\beta = 2$)

k	α	E_A	$\alpha \cdot E_A$	β	E_B	$\beta \cdot E_B$	$\alpha \cdot E_A + \beta \cdot E_B$
1	1	.00	.00	2	.50	1.00	1.00
2	1	.01	.01	2	.45	.90	.91
3	1	.06	.06	2	.33	.66	.72
4	1	.17	.17	2	.17	.34	.51
5	1	.33	.33	2	.06	.12	.45
6	1	.45	.45	2	.01	.02	.47
7	1	.50	.50	2	.00	.00	.50

3. These two groups have opposed preferences for all issues within the class to be considered.

If we consider a "random" individual, one with q/n chances of belonging to the standing majority, it is obvious that majority rule is at least as desirable as any alternative. This, however, avoids the problem, so we must consider two discrete individuals, one a member of the standing majority and one a member of the standing minority. As Figure 3 suggests, there are only three classes of decision-rules which need concern either of these individuals: (1) those which require no more votes than the number commanded by the minority [$k \leq (n - q)$]; (2) those which require more votes than the number commanded by the minority but no more than are commanded by the majority [$k > (n - q), k \leq q$]; and (3) those which require more votes than are commanded by the majority ($k > q$). Given our assumption that the members of these two factions disagree for all proposals within the class considered (3, above), we have only these three kinds of rules to consider.

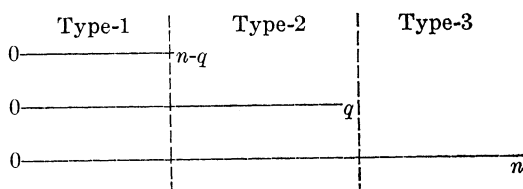


FIG. 3. Three classes of decision-rules for a committee with exactly two consistently opposed factions.

For a member of the standing majority, it is obvious that Type-2 decision-rules are preferable to types 1 and 3. Under Type-2 rules—majority-rule is an example—this member will expect to have his way on all issues belonging to the class considered here. This is because the majority faction can impose all proposals its members like ($E_A = 0$), and veto all proposals its members dislike ($E_B = 0$). Under Type-3 rules, however, he must expect not to have his way on proposals which he and his majority fellows like, for the opposed minority will surely veto them ($E_A =$ the proportion of proposals he likes). And under Type-1 rules, he must expect not to have his way on proposals he dislikes, since the minority will certainly impose them ($E_B =$ the proportion of proposals he dislikes). We would therefore expect the majority member to prefer decision-rules of Type-2—majority-rule, for example.

In contrast, we must expect a minority member to prefer either Type-1 or Type-3 rules, and not Type-2 rules. This is obvious, since Type-1 allows him to impose policies he likes ($E_A = 0$),

and Type-3 rules allow him to veto policies he dislikes ($E_B = 0$), while Type-2 rules allow him neither of these opportunities. And, given our assumptions, there is no possibility of compromise—save the adoption of different rules for various sub-classes of issues.

This bald example of the problem raised by standing majorities and minorities leads to the conjecture that *the problem of the exploited faction is not a special difficulty of majority-rule, but a generic difficulty of all simple-decision-rules*. This is so because *any* rule will leave either the minority or the majority subject to one of three unhappy prospects: the unchecked imposition of unwanted policies; or the invariable veto of wanted policies; or both. It may, of course, be said that Type-1 and Type-3 rules are less open licenses to exploitation than Type-2 rules, since they at least open the prospect of strategic bargaining by the minority. Yet it is hard to imagine the majority which would willingly accept such rules.

Even this much-too-brief consideration of factional conflict suggests that the effort to optimize individual values in collective choice must, in such instances, look beyond the choice of simple decision-rules. No such rule will produce a joint optimization (as did majority-rule under the simple model). Separatism—be it complete, as with the Pakistani withdrawal from India, or partial, as in federal and confederate constitutions—may well be the only prospect for a joint optimization through constitutional choice. Still, separation itself raises the serious prospect that neither faction will, in the long run, fare very well.³¹ Admitting that this consideration of the problem is far from exhaustive, I suspect that one must look away from formal institutional choices and toward the standard civic virtues if individual values are to be optimized in factional strife.

IV. CONCLUSIONS

We arrive at three main conclusions. First, majority rule can be expected to optimize the correspondence between individual values and collective choice under the assumptions of our

³¹ In the terms of this analysis, we might say that partition introduces a special class of events in which policies which Ego favors, and which would have been adopted before the partition, are rejected because the required resources are no longer available. These events are analogous to Event A—the rejection of proposals which Ego favors. These events are singularly relevant to proposals for separation of minorities, as with French Quebec or Black America, since the minority communities would operate with vastly restricted resources.

simple model. Only under a model of this kind can we expect unanimous *a priori* agreement on decision-rules, for the introduction of other information produces determinate conflicts of interests—as illustrated by our second and third conclusions. The second conclusion is that positional preferences constitute a logically legitimate basis for the choice of non-majoritarian decision-rules. It follows that, if there is a conflict over positional preferences, there must also be a conflict over decision-rules, even if other assumptions are unaltered. Third, if we consider the “bald case of factional conflict,” it appears that the problem of the exploited faction is general to all simple decision-rules, and not a special property of majority rule. In this case, we cannot expect factions to agree on any given decision-rule.

These conclusions lead to one last conjecture: a constitutional choice can be made consensually only if uncertainty about the future makes appropriate a model analogous to our simple one. Once decision-rules are conceived as instruments in a determinate conflict of interests, then the incentive for agreement on majority-rule (or any alternative to it) is destroyed, and unanimous consent seems improbable. There may be many ways to obtain agreement on rules in the face of disagreement on issues, but a self-conscious ignorance of the future is probably not the least important among them.

APPENDIX

A mathematical proof devised by Michael Taylor demonstrates that the outcome of the simple models—the optimality of majority-rule—depends only on Assumption I and not on Assumptions II and III.³² This means that the recommendation is general to *all* probabilities for yes votes, regardless of whether or not we take “non-proposals” opposed by all members into account. In effect, Taylor’s proof suggests that Assumptions II and III are theoretically trivial, having simplicity of exposition as their sole purpose. This may be illustrated here by giving a few computational examples which violate Assumptions II and III but nevertheless lead to confirmations of Propositions 1, 2, and 3 (the optimality of majority-rule). In each, we disregard Assumption III and give a specific alternative to Assumption II.

example one:

Consider the following alternative to Assumption II:

$$P_x = \frac{1}{3}, P_x = \frac{2}{3}.$$

³² Taylor, *op. cit.*

Under our new assumptions, there are eight possible support sets in a committee of three (one with three members, three with two, three with one, and one empty). Using the probabilities assigned under this alternative we calculate:

$$P_{xyz} = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$$

$$P_{yz} = P_{xz} = P_{xy} = \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{27}$$

$$P_x = P_y = P_z = \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{4}{27}$$

$$P_{empty} = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{27}$$

$$\left[\frac{1}{27} + \frac{(3 \times 2)}{27} + \frac{(3 \times 4)}{27} + \frac{8}{27} = 1 \right]$$

As before, we may define Events A and B as sets of support sets, and, letting X (or Y or Z) represent Ego, we may calculate the expected frequencies for these events:

Decision-Rule (k)	Expected Frequency Event A	Expected Frequency Event B	Sum of A & B
1	0	$\frac{10}{27}$	$\frac{10}{27}$
2	$\frac{4}{27}$	$\frac{2}{27}$	$\frac{6}{27}$
3	$\frac{8}{27}$	0	$\frac{8}{27}$

Notice that the summary frequency for the two events is no longer symmetrical, but, more important, notice that the minimum for that curve remains at $k=2$, and our finding is unaltered. This is not changed if we reverse the probability assumption, saying that members are twice as apt to support as to oppose each proposal; the curves rotate 180°, but the minimum has the same value and the same location. Or, if we return to Assumption III, barring empty support sets from occurrence, the divisor in each fraction becomes 19 instead of 27, but the location of the A+B minimum is unaltered.

example two:

Now consider an extreme case, in which each of three members is nine times as apt to support as to oppose any proposal:

$$P_x = \frac{9}{10}, P_x = \frac{1}{10}.$$

This activist committee produces support sets with the following probabilities:

$$P_{xyz} = \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} = \frac{729}{1000}$$

$$\begin{aligned} P_{XY} &= P_{XZ} = P_{YZ} = \frac{9}{10} \times \frac{9}{10} \times \frac{1}{10} = \frac{81}{1000} \\ P_X &= P_Y = P_Z = \frac{9}{10} \times \frac{1}{10} \times \frac{1}{10} = \frac{9}{1000} \\ P_{empty} &= \frac{1}{10} \times \frac{1}{10} \times \frac{1}{10} = \frac{1}{1000} \\ \left[\frac{729}{1000} + \frac{(3 \times 81)}{1000} + \frac{(3 \times 9)}{1000} + \frac{1}{1000} = 1 \right] \end{aligned}$$

Expected Frequencies for Events A and B may be computed from these probabilities under our set definitions, and the result is:

Decision-Rule (<i>k</i>)	Expected Frequency Event A
1	0
2	$\frac{999}{1,000,000,000}$
3	$\frac{1,997,001}{1,000,000,000}$

$$\begin{aligned} P_{XYZ} &= \frac{997,002,999}{1,000,000,000} \\ P_{XY} &= P_{XZ} = P_{YZ} = \frac{998,001}{1,000,000,000} \\ P_X &= P_Y = P_Z = \frac{999}{1,000,000,000} \\ P_{empty} &= \frac{1}{1,000,000,000} \\ \left[\frac{997,002,999}{1,000,000,000} + \frac{(3 \times 998,001)}{1,000,000,000} + \frac{(3 \times 999)}{1,000,000,000} + \frac{1}{1,000,000,000} = 1 \right] \end{aligned}$$

Expected Frequency Event B	Sum of A & B
$\frac{999,999}{1,000,000,000}$	$\frac{999,999}{1,000,000,000}$
$\frac{998,001}{1,000,000,000}$	$\frac{999,000}{1,000,000,000}$
0	$\frac{1,997,001}{1,000,000,000}$

Decision-Rule (<i>k</i>)	Expected Frequency Event A	Expected Frequency Event B	Sum of A & B
1	0	$\frac{99}{1000}$	$\frac{99}{1000}$
2	$\frac{9}{1000}$	$\frac{81}{1000}$	$\frac{90}{1000}$
3	$\frac{171}{1000}$	0	$\frac{171}{1000}$

The minimum expected frequency for Events A and B still falls at $k=2$. Admittedly, under these extreme assumptions, it is only slightly lower than the value for $k=1$, but its location is nevertheless stationary. Notice too, that the range of values for this summed frequency is quite low for all three values of k : our activist assumption has placed most of the members on the winning side of most issues (i.e., in 730 of 1000 instances, all three would vote together). Turning this activist committee into a meeting of moss-backs (i.e., reversing the assumption) simply reverses the curves, leaving their shapes and the location of the A+B minimum where it was under Assumption II. And returning to Assumption III simply reduces the denominator in each fraction to 999 instead of 1000.

example three:

As a last example, let us consider an absurd extreme, in which members support proposals with a probability of .999, and oppose them with a probability of .001. Here, we obtain:

It should be clear from these examples that the optimum is stable, at least for very small committees. Taylor's forthcoming proof generalizes this stability to all alternatives for Assumptions II and III, for committees of all sizes. It is quite clear that the degree to which majoritarian decision-rules are superior to their alternatives declines with size³³ and with the asymmetry of the probability assumption, but the nominal fact of their optimality remains in all cases.

³³ This suggests an incentive for the formation of minimum winning coalitions, even if we assume that preferences are independently arrived at so that there is no *a priori* basis for group cohesion. This is because the expected frequency for events A and B is lower, for any given k , in small committees than in large committees. By forming a coalition of k members, and operating under the analogous decision-rule internally, members can always have their way slightly more often than they would by voting independently. In a committee of five, with independent voting, members can expect to have their way 11/16ths of the time if $k=3$; by forming a coalition of three, however, it is possible to raise this expectation to 12/16ths, if $k=2$ is adopted by the coalition. This suggestion is not taken into account by our simple model, which is based on sincere individual voting. This suggestion is an interesting parallel to William Riker's prediction of minimum winning coalitions on the basis of a quite different model. See *The Theory of Political Coalitions* (New Haven: Yale University Press, 1962.)